

Tarea Sección 3.1

$$7) f(t) = 2t^3 - 3t^2 - 4t$$

$$f'(t) = 6t^2 - 6t - 4$$

$$11) y = x^{-2/5}$$

$$y' = -\frac{2}{5}x^{-7/5}$$

$$17) S(p) = \sqrt{p} - p$$

$$S(p) = p^{1/2} - p$$

$$S'(p) = \frac{1}{2}p^{-1/2} - 1$$

$$23) y = \frac{x^2 + 4x + 3}{\sqrt{x}}$$

$$y' = \frac{(2x+4)\sqrt{x} - (x^2+4x+3)\frac{1}{2\sqrt{x}}}{\sqrt{x}^2} =$$

$$y' = \frac{3x^2 + 4x - 3}{2x\sqrt{x}}$$

$$29) f(v) = \frac{\sqrt[3]{v}}{v} - 2ve^v$$

$$f'(v) = \left(\frac{\sqrt[3]{v}}{v} - 2ve^v \right)' = \left(\frac{v^{1/3}}{v} - 2ve^v \right)' = \left(\frac{v^{1/3}(1 - 2v^{2/3}e^v)}{v} \right)'$$

$$= \left(\frac{1 - 2v^{2/3}e^v}{v^{2/3}} \right)' = \left(\frac{1}{v^{2/3}} - 2v^{2/3}e^v \right)' = \left(\frac{1}{v^{2/3}} - 2e^v \right)$$

$$= \left(\frac{1}{v^{2/3}} \right)' + (-2e^v)' = \frac{\frac{2}{3}v^{-5/3}}{(v^{2/3})^2} - 2e^v = \frac{-2}{3v^{7/3}} - 2e^v$$

$$35) y = x + \frac{2}{x} \quad (2, 3)$$

$$y' = 1 - 2\frac{x}{x^2} = y' = 1 - \frac{2}{x^2} \quad m = 1 - \frac{2}{2^2} = \frac{1}{2}$$

$$y - 3 = \frac{1}{2}(x - 2)$$

$$y = \frac{1}{2}x - 1 + 3$$

$$y = \frac{1}{2}x + 2$$

$$37) y = x^2 - x^4 \quad (1, 0)$$

$$y' = 2x - 4x^3$$

$$m = 2(1) - 4(1)^3 = -2$$

tan

$$y - 0 = -2(x - 1)$$

$$y = -2x + 2$$

normal = $\frac{1}{2}$

$$y - 0 = \frac{1}{2}(x - 1)$$

$$y = \frac{1}{2}x - \frac{1}{2}$$

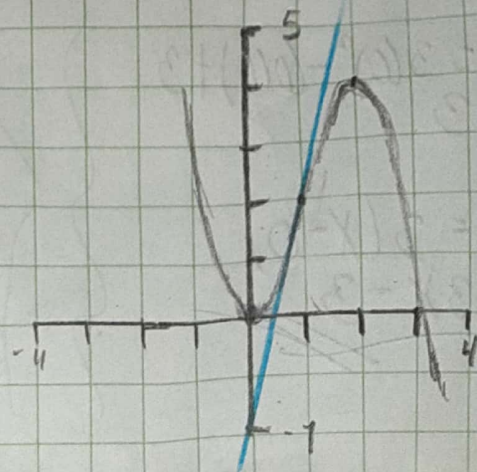
$$39) y = 3x^2 - x^3 \quad (1, 2)$$

$$y' = 6x - 3x^2$$

$$y - 2 = 3(x - 1)$$

$$y = 3x - 1$$

$$y'|_{x=1} = 6(1) - 3(1)^2 = 3$$



Scribe

$$55) y = 2x^3 + 3x^2 - 72x + 7$$

$$y' = 6x^2 + 6x - 72 + 0$$

$$y' = 6x^2 + 6x - 72$$

Para saber en qué puntos es horizontal =

$$y' = 0$$

$$6x^2 + 6x - 72 = 0$$

Cuadrática =

$$x = -2$$

$$x = 7$$

Puntos horizontal = $(-2, 21)(7, -6)$

$$y = 2(-2)^3 + 3(-2)^2 - 72(-2) + 7$$

$$y = -16 + 12 + 144 + 7$$

$$y = 21$$

$$y = 2(7)^3 + 3(7)^2 - 72(7) + 7$$

$$y = 2 + 3 - 144 + 7$$

$$y = -6$$

$$59) y = x^3 - 3x^2 + 3x - 3$$

$$y' = 3x^2 - 6x + 3$$

$$3x - y = 75$$

$$3x^2 - 6x + 3 = 3$$

Cuadrática =

$$x = 0$$

$$x = 2$$

$$y = 0^3 - 3(0)^2 + 3(0) - 3$$

$$y = -3$$

$$P(0, -3)$$

$$y'|_0 = 3(0)^2 - 6(0) + 3$$

$$m = 3$$

$$y + 3 = 3(x - 0)$$

$$y = 3x - 3$$

$$y = 2^3 - 3(2)^2 + 3(2) - 3$$

$$y = -1$$

$$P(2, -1)$$

$$y'|_2 = 3(2)^2 - 6(2) + 3$$

$$m = 3$$

$$y + 1 = 3(x - 2)$$

$$y = 3x - 6 - 1$$

$$y = 3x - 7$$